

Experiment 5

Introduction to Jenkins: What is Jenkins? Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use

1. What is Jenkins?

Jenkins is an open-source automation server used for:

- ✓ Continuous Integration (CI) – Automatically testing and integrating code changes
- ✓ Continuous Deployment (CD) – Automating application deployment
- ✓ Building Pipelines – Managing end-to-end software development workflows
- ✓ Plugin-Based Extensibility – Supporting tools like Maven, Gradle, Ansible, Docker, and Azure DevOps

Why Use Jenkins?

- ✓ Automates builds and tests
- ✓ Reduces manual intervention
- ✓ Improves software quality
- ✓ Works with multiple tools and platforms

2. Installing Jenkins

Jenkins can be installed using multiple methods:

- 1 Windows Installer (.msi) – Recommended for Windows
- 2 Linux Package Manager – Best for Linux Users
- 3 **Jenkins WAR File – Universal method using Java (will be installing by this method)**

● 2.1 Prerequisites

- ◆ Java 21 or 25 is required for Jenkins(from march 31/03/2026 Java 17 will be stoped).

Check Java version:

java -version

● 2.2 Installing Jenkins on Windows

This method allows you to run Jenkins without installing it as a service.

- ✓ Step 1: Download the **Jenkins WAR File**

 Download from: Download and deploy

Choose Generic Java Package (.war).

- ✓ Step 2: Run Jenkins Using Java

Navigate to the folder where the .war file is downloaded and run:

java -jar jenkins.war

- ✓ Step 3: Open Jenkins in Browser

Go to:

<http://localhost:8080>

✅ Step 4: Unlock Jenkins & Setup

Follow the steps as the Windows installation:

✓ Find the **initial password**

✓ Install plugins

✓ Create an admin user

- ◆ Jenkins is now running without installation!

To stop Jenkins, press CTRL + C in the terminal.

◆ **3. Configuring Jenkins for First Use**

Understanding the Jenkins Dashboard

After logging in, you will see:

- ◆ New Item → Create Jobs/Pipelines
- ◆ Manage Jenkins → Configure System, Users, and Plugins
- ◆ Build History → View previous builds
- ◆ Credentials → Store secure authentication details

OUTPUT:



Jenkins / Nitin



Nitin



Profile

Jenkins User ID: nbheemalli



Builds



My Views



Account



Appearance



Preferences



Security



Experiments



Credentials